WORKED SOLUTION

STEP II 2019, Question 11

Probability and Statistics • Counting Triangles

STEP II 2019, Question 11

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

Three lengths form a triangle of non-zero area exactly when the two shorter ones add up to more than the longest. So everything reduces to counting pairs below a fixed largest value, and the cleanest way to count is to fix the middle number n_2 and ask how many values of n_1 are allowed. Part (i) does the counting; parts (ii) and (iii) just divide by the right number of equally likely choices.

Solution

Part (i)

Write $m = n_3$. Fix $n_2 = k$, with $k \leq m-1$. Then n_1 must satisfy

$$m - k < n_1 < k,$$

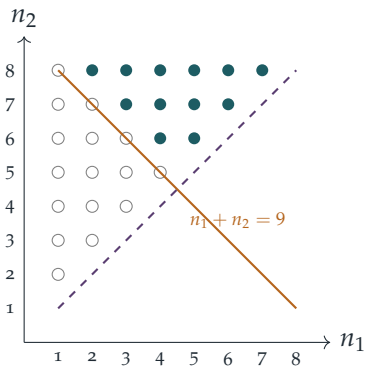
so n_1 runs over $m-k+1, \ldots, k-1$, giving $2k-m-1$ choices whenever this is positive. (The condition $n_1 > 0$ is automatic since $m - k \geq 1$.)

For $m = 9$ the count is $2k - 10$, positive for $k = 6, 7, 8$:

$$2 + 4 + 6 = 12.$$

For $m = 10$ the count is $2k - 11$, positive for $k = 6, 7, 8, 9$:

$$1 + 3 + 5 + 7 = 16.$$



The figure shows the case $n_3 = 9$: the filled dots lie above the line $n_1 + n_2 = 9$ and above the diagonal $n_1 = n_2$, and there are $2 + 4 + 6 = 12$ of them, one column of the triangle for each n_2.

Odd case, $n_3 = 2n + 1$**.** The count for $n_2 = k$ is $2k - 2n - 2 = 2(k - n - 1)$, positive for $k = n + 2, \ldots, 2n$. Putting $j = k - n - 1$, which runs from 1 to $n - 1$,

$$\sum_{j=1}^{n-1} 2j = (n-1)n = n(n-1).$$

Even case, $n_3 = 2n$**.** The count for $n_2 = k$ is $2k - 2n - 1$, positive for $k = n + 1, \ldots, 2n - 1$. With $j = k - n$ running from 1 to $n - 1$,

$$\sum_{j=1}^{n-1} (2j - 1) = (n-1)^2,$$

the sum of the first $n - 1$ odd numbers. Both formulas agree with $12 = 4 \cdot 3$ and $16 = 4^2$.

Part (ii)

The rod of length N is the longest of the three, so the rods form a triangle of non-zero area if and only if the two chosen lengths $a < b$ satisfy $a + b > N$. (The other two triangle inequalities, such as $a + N > b$, hold automatically because $N > b$.) So the favourable choices are exactly the pairs counted in part (i) with $n_3 = N$.

The two rods are chosen from the $N - 1$ remaining, so there are $\binom{N-1}{2}$ equally likely choices.

For $N = 2n + 1$: there are $\binom{2n}{2} = n(2n - 1)$ choices, so

$$\mathrm{P}(\text{triangle}) = \frac{n(n-1)}{n(2n-1)} = \frac{n-1}{2n-1},$$

as required.

For $N = 2n$: there are $\binom{2n-1}{2} = (2n-1)(n-1)$ choices. For this to be non-zero we need $n \geq 2$ (with $N = 2$ there is only one other rod, so two cannot be chosen). Then

$$\mathrm{P}(\text{triangle}) = \frac{(n-1)^2}{(2n-1)(n-1)} = \frac{n-1}{2n-1}.$$

Rather surprisingly, the answer is the same formula as in the odd case, even though N is one smaller.

Part (iii)

Any three distinct rods have a longest one, of length m say, with $3 \leq m \leq 2M+1$. The choice forms a triangle exactly when the other two lengths form one of the pairs counted in part (i) with $n_3 = m$. Each favourable choice is counted once, under its longest rod, so the number of favourable choices is

$$T = \sum_{m=2}^{2M+1} c(m),$$

where $c(m)$ is the count from part (i). (Including $m = 2$ costs nothing, since $c(2) = 0$.)

Ask yourself How can we add up $c(m)$ when it has a different formula for odd and even m?

Pair them up. For each $n = 1, \ldots, M$ take the even value $m = 2n$ and the odd value $m = 2n+1$ together; together these cover $m = 2, 3, \ldots, 2M+1$ exactly once. Their combined contribution is

$$(n-1)^2 + n(n-1) = (n-1)(2n-1) = 2n^2 - 3n + 1.$$

Using the given formula for $\sum k^2$ and $\sum k = \frac{1}{2}M(M+1)$,

$$\begin{aligned} T &= \sum_{n=1}^{M} \left(2n^2 - 3n + 1\right) \\ &= \tfrac{1}{3}M(M+1)(2M+1) - \tfrac{3}{2}M(M+1) + M \\ &= \tfrac{1}{6}M\left[2(2M^2 + 3M + 1) - 9(M+1) + 6\right] \\ &= \tfrac{1}{6}M\left(4M^2 - 3M - 1\right) \\ &= \tfrac{1}{6}M(4M+1)(M-1). \end{aligned}$$

The total number of equally likely choices of three rods is

$$\binom{2M+1}{3} = \frac{(2M+1)(2M)(2M-1)}{6}.$$

Dividing, and cancelling $M/6$,

$$\mathrm{P}(\text{triangle}) = \frac{M(4M+1)(M-1)}{2M(2M+1)(2M-1)} = \frac{(4M+1)(M-1)}{2(2M+1)(2M-1)},$$

as required.

Checks

- $M = 1$: the rods are $1, 2, 3$ and $1 + 2 = 3$ gives a degenerate triangle, so the probability should be 0, and the formula gives 0.
- $M = 2$: rods 1 to 5. The triangles are $\{2, 3, 4\}$, $\{2, 4, 5\}$ and $\{3, 4, 5\}$, so the probability is $3/10$; the formula gives $\frac{9 \cdot 1}{2 \cdot 5 \cdot 3} = \frac{3}{10}$.
- As $M \to \infty$ the probability tends to $\frac{4}{8} = \frac{1}{2}$. This matches the continuous version: for three independent uniform lengths on $[0, 1]$, the chance that the two smaller sum to more than the largest is $\frac{1}{2}$.
- All the counts in part (i) and the probabilities in parts (ii) and (iii) were confirmed by a computer check that listed every pair or triple directly, for n up to 39 and M up to 19 (for example, $M = 3$ gives $13/35$).

Common slips

- Allowing $n_1 + n_2 = n_3$: three rods with lengths $2, 3, 5$ lie flat and have zero area, so the inequality must be strict.
- In part (ii), using $\binom{N}{2}$ instead of $\binom{N-1}{2}$: the rod of length N has already been taken.
- In part (iii), summing only the odd values of m (or only the even ones): the longest rod can have any length from 3 to $2M + 1$.

Beyond STEP

Splitting a count according to the largest element, and then summing, is a standard technique in combinatorics, and the same idea gives the continuous answer $\frac{1}{2}$ by integrating over the value of the maximum. Counting lattice points in a region, as in the figure, and comparing with its area is the bridge between discrete and continuous probability. Blitzstein and Hwang, *Introduction to Probability*, has many problems of this kind, including the classic "broken stick" triangle problem.

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